

SemEval System Papers

Anonymous ACL submission

001	Abstract	3.2 Contrastive Learning Framework	034
002	Abstract	We implement a contrastive learning approach using the Multiple Negatives Ranking Loss (MNRL), which can be formulated as:	035
003	1 Introduction		036
004	Introduction		037
005	(Gusfield, 1997)		
006	2 Background	$L = -\log \frac{\exp(\text{sim}(x_i, x_i^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(x_i, x_j)/\tau)} \quad (1)$	038
007	Background	where:	039
008	3 System Overview	• x_i represents the anchor text embedding (post)	040
009	Our system addresses the challenge of retrieving relevant fact checks from a large corpus of 200,000 documents by implementing a fine-tuned contrastive learning approach based on the multilingual E5 model architecture. The system consists of three main components: (1) a text embedding model, (2) a contrastive learning framework, and (3) a retrieval pipeline.	• x_i^+ represents the positive pair embedding (matching fact-check)	041
010		• x_j represents all other samples in the batch (negative pairs)	042
011		• τ is a temperature parameter that controls the sharpness of the distribution	043
012		• $\text{sim}(\cdot, \cdot)$ is the cosine similarity function	044
013			045
014			046
015			047
016			048
017	3.1 Model Architecture	The framework employs in-batch negatives for computational efficiency while maintaining effective contrastive learning. The temperature parameter τ is dynamically adjusted during training to optimize the learning process.	049
018	The core of our system is built upon the ‘intfloat/multilingual-e5-large-instruct’ model, which we enhance with a custom architecture designed specifically for fact-checking retrieval. The model consists of:		050
019			051
020			052
021			053
022			054
023	• A transformer encoder (base E5 model)	3.3 Retrieval Pipeline	055
024	• A mean pooling layer for sentence representation	The retrieval process follows these steps:	056
025			057
026	• A dense projection layer with GELU activation	1. Language Detection: The system first identifies the language of the input post.	058
027			059
028	• L2 normalization for cosine similarity optimization	2. Corpus Filtering: The fact-check corpus is filtered to prioritize documents in the same language as the input post.	060
029			061
030	The model processes both posts and fact checks using language-specific prefixes (‘query:’ and ‘passage:’ respectively) to optimize the semantic matching between them.	3. Embedding Generation: The model generates embeddings for both the input post and the filtered fact checks.	062
031			
032			
033			

4. **Similarity Ranking:** The system computes Pearson cosine similarity between the post and fact check embeddings.

5. **Top-K Selection:** The 10 most similar fact checks are retrieved and ranked.

This pipeline is optimized for both monolingual and cross-lingual retrieval, with special consideration for language-specific nuances in the embedding space.

4 Experimental Setup

4.1 Dataset

The experimental data consists of three main components:

- **Posts:** Social media posts in multiple languages requiring fact-checking
- **Fact Checks:** A corpus of 153,743 fact-checks across 8 languages (Arabic, German, English, French, Malaysian, Portuguese, Spanish, and Thai)
- **Training Pairs:** Annotated pairs of posts and their corresponding relevant fact-checks

We split the training data into training (80%) and validation (20%) sets while maintaining the language distribution. For each post, we concatenated the OCR-extracted text and manual transcription to create a comprehensive text representation. Similarly, for fact checks, we combined the claim and title fields to create rich semantic representations.

4.2 Implementation Details

The system was implemented using the following configuration:

- **Base Model:** multilingual-e5-large-instruct
- **Framework:** SentenceTransformers (v3.4.1)
- **Deep Learning Backend:** PyTorch (v2.5.1)
- **Python Version:** 3.12.7
- **Computing Environment:** macOS 15.1 (24B83)

4.3 Training Configuration

The model was trained with the following hyperparameters:

Parameter	Value
Maximum Sequence Length	512 tokens
Pooling Mode	Mean
Embedding Dimension	384
Batch Size	4
Evaluation Batch Size	16
Gradient Accumulation Steps	2
Learning Rate	2e-5
Temperature	0.05
Mixed Precision Training	Enabled

Table 1: Model training hyperparameters

4.4 Evaluation Metrics

The primary evaluation metric for our system was Success@10 (S@10), which measures the proportion of posts for which the correct fact check appears in the top 10 retrieved results. We evaluated the model’s performance in both monolingual and cross-lingual settings:

- **Monolingual:** Retrieval within the same language
- **Cross-lingual:** Retrieval across different languages

During training, we also monitored:

- Cosine similarity between positive pairs
- Cosine similarity between negative pairs
- Training loss convergence
- Validation performance at regular intervals

5 Results

We evaluate our system’s performance on both the validation and test sets, analyzing the impact of different model configurations and language-specific performance. Our system achieved an average Success@10 score of 0.867 on the official test set, ranking 20th out of 28 published participants.

5.1 Model Configuration Analysis

Table 2 shows the performance comparison of different base models and input prefix configurations on the validation set. The results demonstrate that the multilingual-e5-large-instruct model with appropriate prefixes consistently outperforms other configurations.

5.2 Language-Specific Performance

Table 3 presents the detailed performance breakdown by language on the validation set, showing significant variation across different languages.

Model	Posts Prefix	Facts Prefix	Success@10
multilingual-e5-large-instruct	passage:	query:	0.834
multilingual-e5-large-instruct	–	–	0.821
multilingual-e5-small	–	–	0.795
e5-large-v2	–	–	0.758
modernbert-embed-base	search_query:	search_document:	0.779
paraphrase-multilingual-mpnet-base-v2	–	–	0.736

Table 2: Performance comparison of base models on the validation set. Models are ordered by Success@10 score. Best result is shown in **bold**.

Language	Success@10	Sample Count	Relative Performance
Thai (tha)	0.930	372	+10.9%
French (fra)	0.895	1,277	+7.4%
Spanish (spa)	0.857	4,502	+3.6%
Arabic (ara)	0.830	541	+0.9%
Malaysian (msa)	0.817	849	-0.4%
German (deu)	0.777	533	-4.4%
Portuguese (por)	0.781	2,057	-4.0%
English (eng)	0.766	3,481	-5.5%
Average	0.821	13,612	–

Table 3: Language-specific performance on the validation set. Relative Performance shows the percentage difference from the average Success@10 score. Best performance is shown in **bold**.

5.3 Impact of Input Prefixes

Our experiments revealed that the choice of input prefixes significantly impacts model performance. Using the recommended prefixes (‘passage:’ and ‘query:’) improved the average Success@10 score by 0.013 points (from 0.821 to 0.834) compared to using no prefixes. This improvement was consistent across all languages, with the largest gains observed in:

- English: +0.022 (from 0.766 to 0.788)
- Spanish: +0.012 (from 0.857 to 0.869)
- Portuguese: +0.018 (from 0.781 to 0.799)

These results suggest that appropriate input formatting plays a crucial role in optimizing the model’s cross-lingual understanding and retrieval capabilities.

6 Conclusion

Conclusion

7 Acknowledgments

Acknowledgments

References

Dan Gusfield. 1997. *Algorithms on Strings, Trees and Sequences*. Cambridge University Press, Cambridge, UK.

A Detailed Results by Language

This appendix presents the detailed performance breakdown of our fine-tuned model across different languages in the official test set. As shown in Table 4, the model achieves consistent performance across diverse language families, from Indo-European (English, French, German) to Austronesian (Malaysian) and Tai-Kadai (Thai) languages.

Language	Success@10	Test Set Size
Malaysian (msa)	0.978	–
Thai (tha)	0.940	–
Arabic (ara)	0.912	–
French (fra)	0.894	–
Spanish (spa)	0.866	–
German (deu)	0.858	–
Turkish (tur)	0.828	–
Polish (pol)	0.806	–
English (eng)	0.796	–
Portuguese (por)	0.796	–
Average	0.867	–

Table 4: Official test set results of the fine-tuned multilingual-e5-large-instruct model by language. Languages are ordered by Success@10 score. Best performance is shown in **bold**. The model shows strong cross-lingual generalization, with particularly strong performance on Malaysian and Thai.